First Data®

BUYPASS® InComm Market Basket Data Format

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Version 2023-1

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Revision History

This is a new document.

The following table lists added, updated, and deleted pages:

Note: Red text and change bars in the page margin indicate updated/added text. Deleted text is indicated only in the following table. Blue text indicates a link to another location in the text.

Changes Made to Ver. No. NA And Reflected in Ver. No. 2023-1 Released August 18, 2023			
Topic	**NA P. No.**	**2023-1 P. No.**	**Summary of Change**
NA	NA	NA	Created the document

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1. Introduction

Market Basket Data should be formatted according to these rules adapted from the *InComm ISO 8583 Generic Interface Specification*, Version 6.2.3, Appendix B. Note that when this document refers to the transaction type, it is referring to the 3-digit value entered as part of the Merchant Supplementary Data. The allowed values are:

Value	Description
310	Benefit Eligibility & Balance Lookup
910	Item Qualification
920	Limited Spend
200	Limited Spend Refund

1.1 Dataset Format

Datasets are composite data elements. Each dataset is formatted in a modified TLV method, with the following format:

Element	Description
TT	A two-character Tag to identify the dataset, such as DV for “Data Version”.
LLL	A three-digit numeric Length (up to “999” in ASCII) to indicate the Length of the dataset. This Length includes all primary and continuation elements for the dataset, as well as the associated bitmaps, but does not include the length of the Tag or Length elements.
BBBBBBBB	An eight-byte Bitmap for the dataset. Dataset bitmaps are implemented as 32-bit bitmaps, represented by eight bytes of ASCII data.
VVV…	The Value of the dataset; that is, all the fields that comprise the dataset.

For example, consider a DV dataset with a binary bitmap. The dataset contains only Bit 2 with the value “01.01”. In hexadecimal, the dataset would be represented as:

44 56 30 30 39 34 30 30 30 30 30 30 30 30 31 2E 30 31 In ASCII, this would be:

“DV0134000000001.01”

1.2 Defined Datasets

There are multiple datasets defined for InComm Healthcare (IH):

Tag	Description
DV	The Data Version indicates which version of a dataset (and its applicable files) is being used. This is for support of future

Tag	Description
	enhancements to the Data Version.
PI	A Purchase Item dataset indicates shared values among a quantity of the same product.
PU	A Purse dataset indicates each active purse associated with a cardholder's account.
OF	An Offer dataset contains the offers available to a cardholder (in a response from InComm), or the offers redeemed (in a request message to InComm). This dataset is restricted to only those retailers who implement an IH solution that does not exchange a market basket in real time.
OC	An Offer Counts dataset contains the information necessary to track the usage of offers that have defined limitations and/or incremental adjustments towards earning an offer. This offer is restricted to only those retailers who implement an IH solution that does not exchange a market basket in real time.

1.3 DV Dataset

The DV dataset is required for all new implementations of IH. It has the following elements:

Bit	Req	Usage	Format	Description
1	O	Continuation Bit	b 1	Indicates a secondary bitmap. (Not currently used)
2	R	Dataset Version	an 5	The dataset version expressed as MM.mm where MM represents the Major version and mm represents the minor version. An InComm integration project resource will provide the value.
3	C	Offer File Version	n 6	This value is used to ensure an appropriate Offer File reference. Required when an implementation does not exchange a market basket in real time, and not applicable otherwise.
4	C	Purse File Version	n 6	This value is used to ensure appropriate Purse File reference.

Bit	Req	Usage	Format	Description
				Required when an implementation does not exchange a market basket in real time, and not applicable otherwise.

1.4 PI Dataset

The PI dataset is required when evaluation of the market basket must be completed by IH to authorize a discount, payment, or refund. Requirements per Bit are represented in the Request and Response definitions. A PI has the following elements:

Bit	Usage	Format	Description
1	Continuation Bit	b 1	Indicates a secondary bitmap. (Not currently used)
2	Item Code	n 14	UPC, EAN13 • Does not include check digit. • Code is right justified, zero filled. Suppressed codes must be expanded.
3	Quantity	n 3	Quantity of the item sold. Quantities should be combined into a single PI when the value of Bits 2, 5, 10, 11, 17, and 18 are the same.
4	Sign Indicator	ans 1	This field is required when the Price field is negative. • 'C' indicates Price field is >= 0.00 • 'D' indicates the Price field is < 0.00
5	Price	n 12	Single retail price shared by all items in the PI Dataset net any store and manufacturer discounts and/or coupons. Do not include tax in a 910 request.
6	MBAS Manufacturer Discount Quantity	n 3	Number of times that the IH Manufacturer Discount Amount is to be applied at the POS. Required when an IH Manufacturer Discount is applied.
7	MBAS Manufacturer Discount Amount	n 12	The per item discount amount to be applied at the POS.

Bit	Usage	Format	Description
			The total IH manufacturer discount is equal to Bit 6 multiplied by Bit 7. Required when an IH Manufacturer Discount is applied.
8	Limited Spend Eligibility Flag	ans 1	• N = Not Eligible for Limited Spend • Y = Eligible for Limited Spend Identifies the item as eligible for spend for the associated account.
9	Redemption Request Amount	n 12	Total redemption request amount for this item code (extended price + all applicable taxes, where the extended price = Quantity * unit Price). Price is net of any applicable discounts. Some states tax discounts and some do not. This amount includes any applicable tax for the item based on retailer business rules applied at the POS. Amount in cents (for example, $100 is represented as 000000010000). Required for 920 and 200.
10	Manufacturer Discount Amount	n 12	The total amount of any manufacturer-funded coupons/discounts already applied to the item at the POS. Note that the Price field is already net of this amount. Amount in cents (ex. $100 would be shown as 000000010000). Required when the POS has applied a manufacturer discount.
11	Manufacturer Discount Type	ans 1	The delivery method of the Manufacturer Discount Amount. This field is required if the Manufacturer Discount Amount is present, or when no further Manufacturer Discounts are allowed by POS rule. • P = Paper Manufacturer Coupon • E = Electronic Manufacturer Coupon • M = Mixed (both P and E). • X = Additional Manufacturer Discounts are not allowed.

Bit	Usage	Format	Description
			Required when the POS has applied a manufacturer discount.
12	Adjustment to Price	n 12	The amount of the adjustment, that is, the amount paid by InComm including any multiplying effect of quantity for the PI dataset. For example, an item qualification provides a Price of $5.00 and a Quantity of 2. The discount applied by MBAS is $0.75 for each item, so the value of this field $1.50, the product of 2 * $0.75. Amount in cents (for example, $0.75 is represented as 000000000075). Required when IH has discounted or paid for a portion or all of a PI.
13	Item Reference Number	n 4	PI reference number enables the retailer to match requests to responses in the PI dataset. This number must be unique per PI on the request. The uniqueness applies to the whole message. Item Reference Numbers do not need to be contiguous. PI datasets are not necessarily returned in the order sent to the host. In cases in which a PI dataset is sent with a quantity value greater than one (1), the dataset may be 'split' on the responses. This event can occur if: • Item qualification responses when a discount is eligible only for part of the quantity submitted • Limited spend responses when either a purse balance has been depleted (full payment PI, partial payment PI, not paid PI) • Limited spend responses when multiple purses are used to fund items. When PI datasets are 'split' in this fashion, each of the resultant PI datasets will bear the same Item Reference Number as was contained in the original request.

Bit	Usage	Format	Description
14	Purse ID	n 3	Purse identification for cross-reference to the applicable PU dataset Purse ID. A value of zero (000) in this Bit indicates that the PI is not associated with a PU. The Purse ID on Item Qualification responses is set to zero (000) because the actual association is not known until items are applied to purse balances authorized in Limited Spend. Required on responses containing a purse.
15	MBAS Store Discount Quantity	n 3	Number of times that the IH Store Discount Amount is to be applied at the POS. Required when an IH Store Discount is applied.
16	MBAS Store Discount Amount	n 12	The per item discount amount to be applied at the POS. The total IH store discount is equal to Bit 15 multiplied by Bit 16. Required when an IH Store Discount is applied.
17	Store Discount Amount	n 12	The total amount of any store-funded coupons/discounts already applied to the item at the POS. Note that the Price field is already net of this amount. Amount in cents (ex. $100 would be shown as 000000010000) Required when the POS has applied a store discount.
18	Store Discount Type	ans 1	The delivery method of the Store Discount Amount. This field is required if the store Discount Amount is present, or when no further Store Discounts are allowed by POS rule. • P = Paper Store Coupon • E = Electronic Store Coupon • M = Mixed (both P & E) • X = Additional Store Discounts

<table>
<tr><th>Bit</th><th>Usage</th><th>Format</th><th>Description</th></tr>
<tr><td></td><td></td><td></td><td>are not allowed
Required when the POS has applied a store discount.</td></tr>
<tr><td>19</td><td>MBAS Eligibility Flags</td><td>ans 20</td><td>Eligibility Flags describing the IH discount.
Y = Eligible
N = Not Eligible
<table>
<tr><th>Position</th><th>Eligibility Type</th></tr>
<tr><td>1</td><td>MBAS Manufacturer Discount</td></tr>
<tr><td>2</td><td>MBAS Store Discount</td></tr>
<tr><td>3</td><td></td></tr>
<tr><td>4-20</td><td>Reserved</td></tr>
</table></td></tr>
<tr><td>20-32</td><td>Reserved</td><td></td><td>Reserved for future use</td></tr>
</table>

1.5 PI Dataset Field Usage

A PI dataset is optional in a Benefit Eligibility & Balance Lookup (310). It is not present in implementations that do not exchange a market basket.

This table presents all the PI fields and their requirements per Transaction Type. Rows in orange represent discount information conveyed by the retail partner in the request, and rows in blue represent discount information conveyed by InComm in its response.

		Request Req		Response Req	
Bit	**Usage**	**310 & 910**	**920 & 200**	**310 & 910**	**920 & 200**
2	Item Code	R	R	RE	RE
3	Quantity	C	C	CE*	CE*
4	Sign Indicator	O	NA	OE	NA
5	Price	R	NA	RE	NA
6	MBAS Manufacturer Discount Quantity	NA	NA	C	NA
7	MBAS Manufacturer Discount Amount	NA	NA	C	NA

		Request Req		Response Req	
Bit	**Usage**	**310 & 910**	**920 & 200**	**310 & 910**	**920 & 200**
8	Limited Spend Eligibility Flag	NA	NA	R	R
9	Redemption Request Amount	NA	R	NA	E
10	Manufacturer Discount Amount	C	NA	NA	NA
11	Manufacturer Discount Type	C	NA	NA	NA
12	Adjustment to Price	NA	NA	R**	R
13	Item Reference Number	R	R	RE	RE
14	Purse ID	NA	NA	R	R
15	MBAS Store Discount Quantity	NA	NA	C	NA
16	MBAS Store Discount Amount	NA	NA	C	NA
17	Store Discount Amount	C	NA	NA	NA
18	Store Discount Type	C	NA	NA	NA
19	MBAS Eligibility Flags	NA	NA	R	R

* The quantity can be split up into multiple PIs in the response if IH applies discounts or spend to only a portion of the requested quantity.

** The adjustment to price is not provided on a 310 as this is a preauthorization.

1.6 PU Dataset

Purse Datasets are returned in responses by IH. One PU dataset is returned for each active purse associated with an account. Purses with no balance (e.g., 0.00) remaining are included in the response. It has the following base elements:

Bit	Usage	Format	Description
1	Continuation Bit	b 1	Indicates a secondary bitmap. (Not currently used.)
2	Purse Description	an 20	A label to describe the purse. This value should be presented to the consumer along with the balance at the conclusion of a transaction.
3	Purse ID	n 3	A numeric identifier of the purse.

Bit	Usage	Format	Description
4	Sign Indicator	ans 1	• 'C' indicates the remaining balance Bit is >= 0.00 • 'D' indicates the remaining Balance Bit is < 0.00 This field is required when the Price field is negative.
5	Remaining Balance	n 12	The remaining balance for the purse. Amount in cents (ex. $100 would be shown as 000000010000)
6	Amount Authorized	n 12	The amount authorized from the purse. This amount is never negative. This amount is zero (000000000000) for purse declines or for purses not involved in the transaction (that is, no PI was associated with the purse). Amount in cents (ex. $100 would be shown as 000000010000) Required for 920 and 200
7	Response Code	an 2	The response code may provide additional information about a purse decline.
8	Purse Unique ID	n 10	The unique purse number used for the redemption or refund Applicable only to an implementation that does not exchange a market basket in real time, and not applicable otherwise.
9	Redemption Request Amount	n 12	Total redemption request amount for this purse Applicable only to an implementation that does not exchange a market basket in real time, and not applicable otherwise.
10-32	Reserved		Reserved for future use.

1.7 PU Dataset Field Usage

This table presents all the PU fields and their requirements per applicable Transaction Type.

		Request Req	Response Req			
Bit	**Usage**	**920 & 200**	**310**	**310**	**920 & 200**	**920 & 200**
2	Purse Description	NA	R	R	R	R
3	Purse ID	NA	R	NA	R	NA
4	Sign Indicator	NA	C	NA	C	C
5	Remaining Balance	NA	R	NA	R	R
6	Amount Authorized	NA	NA	NA	R	R
7	Response Code	NA	R	NA	R	R
8	Purse Unique ID	R	NA	R	NA	R
9	Redemption Request Amount	R	NA	NA	NA	R

1.8 OF Dataset

The OF dataset is required for implementations of IH which do not exchange a market basket. It has the following elements:

Bit	Usage	Format	Description
1	Continuation Bit	b 1	Indicates a secondary bitmap. (Not currently used.)
2	Offer ID	n 10	Numerical identifier assigned by InComm to an offer.
3	Offer Tag	ans 4	A label that describes the basic characteristics of an offer.
4	Offer Print Text	ans 25	A short, literal description of an offer. Designed to be printed on the receipt. Examples: Buy3Get1FreeTheraflu $1OffJergens
5	Offer Verbiage	ans 100	A longer, readable version of the Offer Print Text. Examples: Buy Three Theraflu Get One Free Get $1 Off Jergens

Bit	Usage	Format	Description
6-32	Reserved	NA	Reserved for future use.

1.9 OF Dataset Field Usage

This table presents all the OF fields and their requirements per Transaction Type.

		Request Req	Response Req
Bit	**Usage**	**910**	**310**
2	Offer ID	R	R
3	Offer Tag	NA	R
4	Offer Print Text	NA	R
5	Offer Verbiage	NA	R

1.10 OC Dataset

The OC dataset is required for implementations of IH which do not exchange a market basket. It has the following elements:

Bit	Usage	Format	Description
1	Continuation Bit	b 1	Indicates a secondary bitmap. (Not currently used.)
2	Offer ID	n 10	Numerical identifier assigned by InComm to an offer.
3	Times	n 10	Indicates the number of times something has occurred in a count.
4	Amount	n 12	Indicates an amount in pennies of a monetary count.
5	Type	a 20	Indicates the type of count being tracked. In an **Account Lookup** response, the valid types are: **tallyEarn** – indicates to the Retailer the total of purchases towards this accumulated offer (e.g., the consumer has previously bought 3 items towards a Buy 4, Earn $1 Off the Fifth offer) **tallyRedeem** – indicates to the Retailer

Bit	Usage	Format	Description
			the portion already redeemed of a previously earned offer (e.g., the consumer has used $15 of the offer of Buy $100 of One, Get $25 Towards Another) **redeemedCount** – indicates to the Retailer total redemptions by the consumer of this offer (e.g., the consumer has previously redeemed the Buy 4, Earn $1 Off the Fifth offer 4 times) In an Offer Redemption request, the valid types are: **incrementedEarn** – indicates to InComm the number of incremental purchases towards this accumulating offer during the transaction (e.g., the consumer bought 3 items towards a Buy 4, Earn $1 Off the Fifth offer) **incrementedRedeem** – indicates to InComm the redemption of this offer that has a redemption threshold (e.g., The consumer redeemed $15 worth on another item for the offer of Buy $100 of One, Get $25 Towards Another) **timesRedeemed** – indicates to InComm how many times this offer was redeemed during the transaction (e.g., the consumer has redeemed the Buy 4, Earn $1 Off the Fifth offer one time in the transaction) **totalAmountAwarded** – indicates to InComm the total amount of money awarded in the transaction for this offer (e.g., the consumer has redeemed the Buy 4, Earn $1 Off the Fifth offer twice in the transaction for a total amount awarded of $2) **Note: totalAmountAwarded** will need to be converted by the Retailer to an amount when redeeming quantity or percentage-based offers.
6-32	Reserved	NA	Reserved for future use.

1.11 OC Dataset Field Usage

This table presents all the OF fields and their requirements per Transaction Type.

		Request Req	Response Req
Bit	**Usage**	**910**	**310**
2	Offer ID	R	C
3	Times	C	C
4	Amount	C	C
5	Type	C	C